Empowering SFDA via MLLM-Guided Reliability-Based Curriculum Learning

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TL;DR

Multi-teacher MLLM distillation + reliability-based curriculum learning for SOTA Source-Free Domain Adaptation!

Background and Motivation

Problem

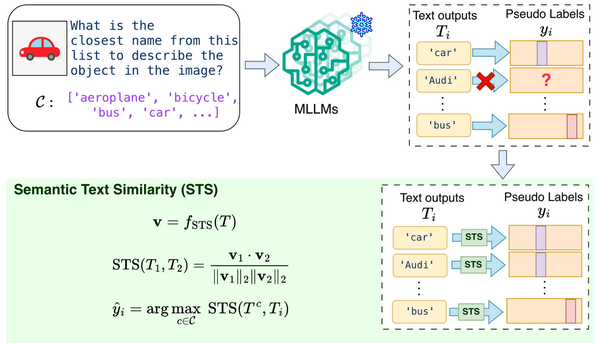
- Source-Free Domain Adaptation (SFDA) adapts models **without access to source data**
- Existing SFDA methods:
  - Single-model pseudo-labels
  - Handcrafted prompts or confidence filtering
- Limited zero-shot performance of multimodal LLMs (MLLMs)
  - Predictions can be inconsistent across samples
  - Outputs are noisy and free-form
  - Direct use is computationally expensive
- Challenge: **use strong teachers while controlling label noise**

Key Idea

- Use **multiple frozen MLLMs** as supervision sources
- Measure **teacher agreement** to estimate pseudo-label **reliability**
- Train progressively using a **reliability-based curriculum**
- Learn a compact student model for efficient deployment

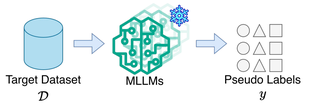
Pseudo-labeling with STS

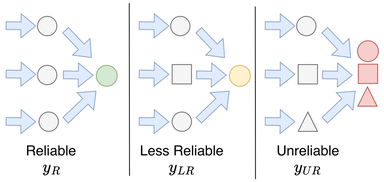
- MLLM outputs are free-form and not label-aligned
- STS maps outputs and classes into a shared embedding space
- Pseudo-labels are assigned via maximum semantic similarity



Reliability-Based Curriculum Learning

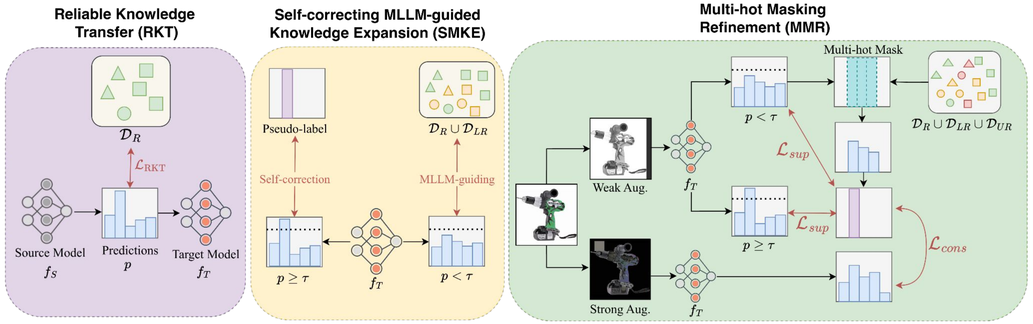
- Multiple frozen MLLMs generate pseudo-labels
- Teacher agreement estimates label reliability
- Dataset split into reliable / less-reliable / unreliable subsets





Reliability-based Curriculum Learning (RCL)

- Reliable (RKT) → Less reliable (SMKE) → Unreliable (MMR)



Main Results

- Accuracy (%) on SFDA (adaptation from Source → Target domain), and Average across all adaptation tasks.
- **State-of-the-Art on Office-Home, DomainNet, VisDA** across all adaptation tasks

Method	Venue	SF	CP	ViT	Office-Home A→C	A→P	A→R	C→A	C→P	C→R	P→A	P→C	P→R	R→A	R→C	R→P	Avg.	DomainNet Avg.	VisDA Avg.
Source	-	-	✗	✗	44.7	64.2	69.4	48.3	57.9	60.3	49.5	40.3	67.2	59.7	45.6	73.0	56.7	52.7	45.3
DAPL-RN	NeurIPS'23	✗	✓	✗	54.1	84.3	84.8	74.4	83.7	85.0	74.5	54.6	84.8	75.2	54.7	83.8	74.5	74.8	86.9
PADCLIP-RN	ICCV'23	✗	✓	✗	57.5	84.0	83.8	77.8	85.5	84.7	76.3	59.2	85.4	78.1	60.2	86.7	76.6	–	88.5
ADCLIP-RN	ICCV'23	✗	✓	✗	55.4	85.2	85.6	76.1	85.8	86.2	76.7	56.1	85.4	76.8	56.1	85.5	75.9	75.2	88.5
PLUE	CVPR'23	✓	✗	✗	49.1	73.5	78.2	62.9	73.5	74.5	62.2	48.3	78.6	68.6	51.8	81.5	66.9	64.7	88.3
C-SFDA	CVPR'23	✓	✗	✗	60.3	80.2	82.9	69.3	80.1	78.8	67.3	58.1	83.4	73.6	61.3	86.3	73.5	–	87.8
PSAT-GDA	TMM'23	✓	✗	✓	73.1	88.1	89.2	82.1	88.8	88.9	83.0	72.0	89.6	83.3	73.7	91.3	83.6	–	86.3
TPDS	IJCV'24	✓	✗	✗	59.3	80.3	82.1	70.6	79.4	80.9	69.8	56.8	82.1	74.5	61.2	85.3	73.5	67.1	87.6
DIFO-C-RN	CVPR'24	✓	✓	✗	62.6	87.5	87.1	79.5	87.9	87.4	78.3	63.4	88.1	80.0	63.3	87.7	79.4	76.7	88.8
DIFO-C-B32	CVPR'24	✓	✓	✓	70.6	90.6	88.8	82.5	90.6	88.8	80.9	70.1	88.9	83.4	70.5	91.2	83.1	80.0	90.3
LCFD-C-RN	–	✓	✓	✗	60.1	85.6	86.2	77.2	86.0	86.3	76.6	61.0	86.5	77.5	61.4	86.2	77.6	78.0	89.3
LCFD-C-B32	–	✓	✓	✓	72.3	89.8	89.9	81.1	90.3	89.5	80.1	71.5	89.8	81.8	72.7	90.4	83.3	80.0	89.3
LLaVA-34B*	NeurIPS'23	-	✓	✓	78.3	93.7	89.5	87.0	93.7	89.5	87.0	78.3	89.5	87.0	78.3	93.7	87.2	86.1	92.1
InstBLIP-XXL*	NeurIPS'23	-	✓	✓	82.0	91.6	88.8	82.2	91.6	88.8	82.2	82.0	88.8	82.2	82.0	91.6	86.2	85.3	86.7
ShrGPT4V-13B*	ECCV'24	-	✓	✓	66.7	85.8	84.8	83.2	85.8	84.8	83.2	66.7	84.8	83.2	66.7	85.8	80.1	81.7	90.4
RCL (Ours)	-	✓	✗	✗	82.5	95.3	**93.3**	89.1	**95.3**	**92.7**	**89.3**	**82.4**	92.8	89.4	82.1	95.4	90.0	89.4	93.2
RCL-ViT (Ours)	-	✓	✗	✓	**83.1**	**95.7**	93.1	**89.2**	**95.3**	92.6	89.2	82.3	**92.9**	**90.0**	**83.2**	**95.5**	**90.2**	**89.7**	**93.3**

SF: Source-Free ; CP: uses CLIP; ViT: ViT backbone; *Zero-shot with STS

Ablation Studies

Component Ablation

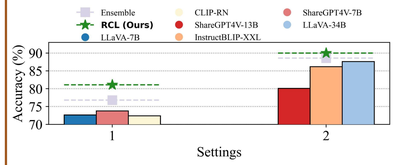
RCL RKT	SMKE	MMR	Office-Home →A	→C	→P	→R	Avg.
✓	✗	✗	82.8	73.3	89.3	88.1	83.3
✓	✗	✓	87.7	80.2	93.3	92.0	88.3
✓	✓	✗	88.5	80.9	95.1	92.5	89.3
✓	✓	✓	**89.3**	**82.3**	**95.3**	**92.9**	**90.0**

RCL Without MLLMs

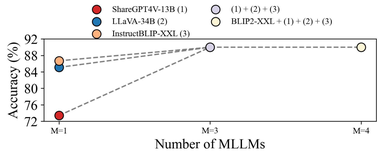
Method	→C	→P	→R	→A	Avg.
TPDS (A)	59.1	81.7	81.7	71.6	73.5
LCFD-C-B32 (B)	72.2	90.2	89.7	81.0	83.3
DIFO-C-B32 (C)	70.4	**90.8**	88.8	**82.3**	83.1
RCL (A, B, C)	**71.9**	90.7	**89.2**	81.7	**83.4**

- Each RCL stage/component improves performance incrementally
- Even without MLLMs, RCL gets competitive performance

Different Teacher MLLMs



Number of Teacher MLLMs



- RCL remains robust across different teacher choices
- Performance improves up to three teachers, and saturates beyond

Backbone Ablation

Method	BB	Office-Home →A	→C	→P	→R	Avg.
DIFO-C-RN	RN50	79.3	63.1	87.7	87.5	79.4
DIFO-C-B32	RN50	82.3	70.4	90.8	88.3	83.1
RCL (Ours)	RN18	89.1	81.5	95.1	92.6	89.6
RCL (Ours)	RN50	**89.3**	**82.3**	**95.3**	**92.9**	**90.0**

Prompt Sensitivity

Prompt Template	A→C	A→P	A→R	Avg.
Naive prompt	76.80	92.10	87.45	85.71
Default prompt (D)	**78.35**	**93.78**	**89.58**	**87.19**
D + Domain info	77.54	93.08	88.73	86.25
D + Paraphrased classes	76.20	92.45	87.90	85.20
D + Distractor labels	76.90	92.70	88.25	85.79

- Performance gains are architecture-agnostic
- STS reduces sensitivity to prompt variations for teacher pseudo-label generation

Conclusion / Summary

- Introduce MLLMs as foundation teachers for SFDA task
- Align free-form MLLM outputs using STS
- Propose Multi-teacher consensus to estimate label reliability
- Design RCL, a three-stage curriculum learning framework:
  - Reliable → Less reliable → Unreliable samples
- Achieves SOTA SFDA without foundation model fine-tuning

RCL improves robustness by prioritizing reliable supervision before learning from noisy samples.

Project Page

Code

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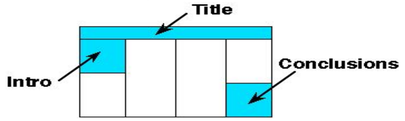
Abstract

Section 1 (sizes):

- Posters should be 24" tall and 48" wide, but we recommend you leave a little border since you may not be able to pin at the vertical edge. Since PowerPoint does not let one define such a large paper size, this template is designed to be printed at 200%, yielding a 46" x94" poster. You can scale it up or down a bit (e.g. 42" is a common paper size at FexEd). Note there is no direct international A0.. A1 equivalent. The poster size is approximately three A0 boards next to each other, i.e., each column in this example is about one A0 board.

- Ideally you want to keep it very readable: this is not your paper, it is a poster. 32pt here (64 final printing) is good for most text:
  - Sub-bullets are 28 here (56 final)
  - Don't use smaller than 24pt in this template (which is 48pt in final printing at 200%)
  - Insert plenty of graphics and any math you need

- When inserting graphics or equations, keep the resolution high (remember this will be printed at 200%). If you can see blocking artifacts at 400% magnification in PowerPoint, consider finding better graphics. This is an example of BAD/LOW RES GRAPHICS



- Leave enough margin for pushpin and remember many big plotters cannot get within .5" of the actual paper edge.

- You are free to use colored backgrounds and such but they generally reduce readability.

- You are free to use what ever fonts you like.
  - San Serif fonts like Arial are more readable from a distance,
  - Serif fonts like times may look more consistent with your mathematics

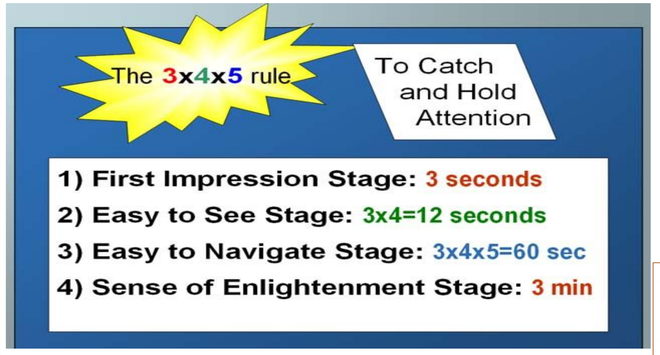
Methods and Materials

Section 2 (layout):

- Remember the poster session will be crowded so design the poster to be read in columns so people can read what is in front of them and move left to right to get the whole story.

- The poster should use photos, figures, and tables to tell the story of the study. For clarity, present the information in a sequence that is easy to follow.

- There is often way too much text in a poster - there definitely is in this template! Posters primarily are visual presentations; the text should support the graphics. Look critically at the layout. Some poster 'experts' suggest that if there is about 20-25% text, 40-45% graphics and 30-40% empty space, you are doing well



Results

Section 3:

- Include more figures than are in the paper so you can talk to them. Include things that are not in the paper and then encourage them to read the paper. Don't try to just put all the paper here.

- If it looks like a cut/paste of the paper, people skip that poster since they can read the papers after the conference. Many people find it better to spend time talking with poster presenters that have more to offer than just redoing the paper content paper in big fonts.

- People will likely have already seen your posted video, so the poster can serve as a talking point for you.

- Remember Poster boards look like this (you will be sharing this space with another poster.. This is your canvas. Paint us a picture of your work.

Click here to insert more Results text. Type it in or copy and paste from your Word document or other source.

To change the font style of this text box: Click on the border once to highlight the entire text box, then select a different font or font size that suits you. This text is Calibri 24pt and is easily read up to 4 feet away on a 24x48 poster, up to 6 feet away on a 36x72 poster, and up to 8 feet away on a 48x96 poster.

Zoom out to 100% (for 24x48), 150% (for 36x72), or 200% (for 48x96) to preview what this will look like on your printed poster.

Discussion

Conclusions

Summary/Conclusion

- Summarize your contributions

- Summarize your results (if applicable)

- You an add in links to additional videos, code, or project website (or QR code)

References

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